

Chemistry Ch 3.7: Electronic Configuration

Chapter 3, Atomic Structure · kitab ke page 27 se 44

Shells aur sub-shells

Shells ko harfon se likhte hain: **K = 1, L = 2, M = 3, N = 4.**

Har shell ke andar **sub-shells** hoti hain, aur har ek ki apni gunjaish hai:

s = 2 electrons · p = 6 · d = 10 · f = 14

Auf Bau Principle

Electrons hamesha **sab se kam energy wali sub-shell pehle** bharte hain.

Energy ki tarteeb: **1s < 2s < 2p < 3s < 3p < 4s < 3d ...**

4s pehle, 3d baad mein

Ghor karein: **4s** ka number bara hai magar wo **3d se pehle** aata hai. Ye galti sab se zyada hoti hai.

Wajah: shell ka number nahin, **energy** ahem hai, aur 4s ki energy 3d se kam hai. Tarteeb **ratt lein**.

Likhne ka tareeqa

Sodium (Z = 11), yani 11 electrons: **1s² 2s² 2p⁶ 3s¹**

Ginti kar lein: 2 + 2 + 6 + 1 = 11. Ye ginti **hamesha** karein, yehi check hai.

Notation: mass number **ooper bayein**, atomic number **neeche bayein**, charge **ooper dayein**.

Quiz

1. Name the first four shells with their numbers, and give the maximum electrons each sub-shell can hold.
2. State the **Auf Bau principle**, write the energy order up to 3d, and give the electronic configuration of sodium (Z = 11). Explain why 4s fills before 3d.